

TrES-3b

R-2762

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220812 · created 2026-09-03T00:32:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459803.8198 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.035 +/- 0.057
Transit depth (Rp/Rs)^2	0.12 +/- 0.4 [%]
Orbital Inclination (inc)	77.81 +/- 2.86
Airmass coefficient 1 (a1)	0.94 +/- 0.13
Airmass coefficient 2 (a2)	0.04 +/- 0.29
Scatter in the residuals of the lightcurve fit is	13.37 %
Best Comparison Star	#2 - [223, 412]
Optimal Aperture	3.03
Optimal Annulus	6.77
Transit Duration (day)	0.006 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.035 ± 0.057 vs 0.16309 (deviation 2.25σ)
Duration fit vs published	0.006 d vs 0.0567 d (deviation 3.17σ)
Mid-time O–C	1.3 min
Depth SNR	0.6
Residual scatter	13.37 %

Light curve

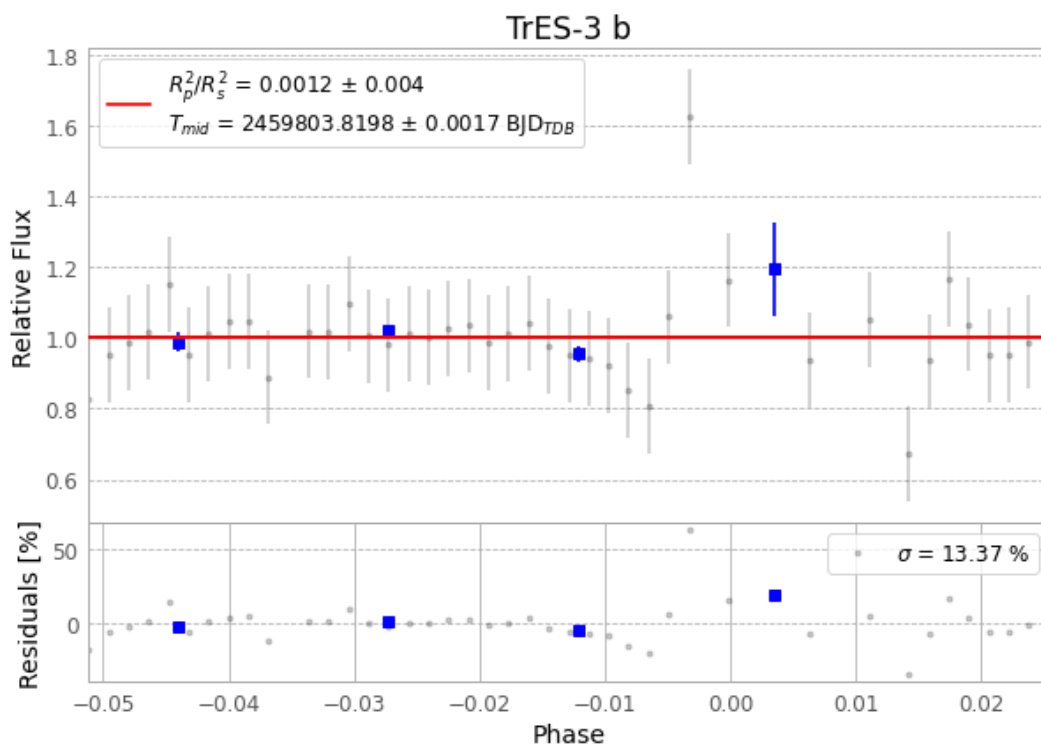


Figure 1 — FinalLightCurve_TrES-3 b_2022-08-11.png (SHA-256 b0f72a5aab899f82...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 328], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fc6e3607b5af50aa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3220812060013.FITS → TRES-3220812091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220812.log (f08114b7c2cb...), FinalLightCurve_TrES-3 b_2022-08-11.png (b0f72a5aab89...), FinalLightCurve_TrES-3 b_2022-08-11.csv (4cf7874b3aa2...)

Compute resources

Wall time 262.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:32Z.